SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	2026 (SLOT A)	**Date:**	21/10/2026

Code of Conduct

I Vijay Dharsith *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula $s = \text{round}((L - 1) \times CDF)$, where $L = 256$. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3×3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3×3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

COMPUTER VISION
ANALYTICAL PROBLEMS – ASSIGNMENT ANSWERS

Q1. Histogram Equalization

Given pixel intensity values: 10, 10, 20, 20, 20, 30, 30, 40.

Number of pixels, N = 8. Number of intensity levels, L = 256.

Step 1: Construct the histogram

Intensity value	Frequency (f)	Cumulative frequency	CDF
10	2	2	2/8 = 0.25
20	3	5	5/8 = 0.625
30	2	7	7/8 = 0.875
40	1	8	8/8 = 1.00
Total	8		

Step 2: Apply histogram equalization

Formula: $s = \text{round}((L - 1) \times \text{CDF})$

Since $L = 256$: $s = \text{round}(255 \times \text{CDF})$

For intensity 10: $s = \text{round}(255 \times 0.25) = \text{round}(63.75) = 64$

For intensity 20: $s = \text{round}(255 \times 0.625) = \text{round}(159.375) = 159$

For intensity 30: $s = \text{round}(255 \times 0.875) = \text{round}(223.125) = 223$

For intensity 40: $s = \text{round}(255 \times 1.00) = 255$

Step 3: New intensity values

Original intensity	Frequency	CDF	New intensity
10	2	0.25	64
20	3	0.625	159
30	2	0.875	223
40	1	1.00	255

The equalized pixel values are: 64, 64, 159, 159, 159, 223, 223, 255.

Comparison before and after equalization

Before equalization, the pixel values lie between 10 and 40, so the intensity range is narrow and contrast may be low. After equalization, the values span from 64 to 255, making differences between intensity levels more noticeable.

Result: Histogram equalization spreads the intensity values over a wider range and can improve image contrast.

Q2. Averaging Filter and Median Filter

Given 3 × 3 image:

10	20	30

20	100	20
10	20	30

The original centre pixel is 100.

Step 1: Apply the averaging filter

Average = sum of all 9 pixel values ÷ 9

Average = (10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30) ÷ 9

Average = 260 ÷ 9 = 28.89 ≈ 29

Replace the centre pixel with 29. Resulting image:

10	20	30
20	29	20
10	20	30

Step 2: Apply the median filter

Sorted values: 10, 10, 20, 20, 20, 20, 30, 30, 100

The middle (5th) value is 20. Therefore, the new centre pixel = 20.

Resulting image:

10	20	30
20	20	20
10	20	30

Step 3: Comparison

Feature	Averaging filter	Median filter
Centre pixel value	29	20
Method	Calculates the mean	Selects the middle value
Noise removal	Affected by extreme values	Effective for isolated extreme values
Edge preservation	May blur edges	Generally preserves edges better

Result: The median filter is better for removing salt-and-pepper noise because the extreme value 100 does not strongly affect the median. The averaging filter is more affected by it.

Q3. Laplacian Filter for Image Sharpening

Given intensity matrix:

50	50	50
50	100	50
50	50	50

Laplacian filter mask:

0	-1	0
-1	4	-1
0	-1	0

The original centre pixel intensity is 100.

Step 1: Calculate the Laplacian response

$$\text{Response} = (0\times50) + (-1\times50) + (0\times50) + (-1\times50) + (4\times100) + (-1\times50) + (0\times50) + (-1\times50) + (0\times50)$$

$$\text{Response} = 0 - 50 + 0 - 50 + 400 - 50 + 0 - 50 + 0 = 200$$

Laplacian response = 200.

Step 2: Sharpen the centre pixel

$$\text{Sharpened intensity} = \text{original intensity} + \text{Laplacian response}$$

$$\text{Sharpened intensity} = 100 + 200 = 300$$

Therefore, the calculated sharpened centre intensity is 300. In an 8-bit grayscale image, values are limited to 0–255, so 300 would be clipped to 255 if stored directly.

Step 3: Explanation

The Laplacian filter detects rapid changes in pixel intensity. It produces a strong response where the centre pixel differs from its neighbours. Adding this response to the original pixel increases the intensity difference and emphasizes edges and fine details.

Result: The Laplacian response is 200, and the sharpened centre intensity is 300 before clipping.

Q4. Global Thresholding

Given grayscale intensity matrix:

40	60	120
80	150	200
30	100	180

Threshold value: T = 100.

Step 1: Apply the threshold

$$\text{If intensity} \geq 100, \text{ assign } 1; \text{ if intensity} < 100, \text{ assign } 0.$$

The values 120, 150, 200, 100 and 180 become 1. The values 40, 60, 80 and 30 become 0.

Step 2: Resulting binary image

0	0	1
0	1	1
0	1	1

Step 3: Count foreground and background pixels

Pixel category	Binary value	Number of pixels
Foreground	1	5
Background	0	4
Total		9

Step 4: Explanation and limitation

Global thresholding separates an image into foreground and background using one threshold. Pixels with intensity greater than or equal to 100 are classified as foreground; the remaining pixels are classified as background.

Limitation: It may not work well when lighting varies across an image, because one threshold can incorrectly classify some foreground and background pixels.

Result: The binary image contains 5 foreground pixels and 4 background pixels.

Q5. Sobel Operator for Edge Detection

Given intensity matrix:

10	10	10
10	50	90
10	50	90

Sobel masks:

Gx =

-1	0	1
-2	0	2
-1	0	1

Gy =

-1	-2	-1
0	0	0
1	2	1

Step 1: Calculate the horizontal gradient Gx

$$Gx = (-1\times10) + (0\times10) + (1\times10) + (-2\times10) + (0\times50) + (2\times90) + (-1\times10) + (0\times50) + (1\times90)$$

$$Gx = -10 + 10 - 20 + 180 - 10 + 90 = 240$$

Step 2: Calculate the vertical gradient Gy

$$Gy = (-1\times10) + (-2\times10) + (-1\times10) + (0\times10) + (0\times50) + (0\times90) + (1\times10) + (2\times50) + (1\times90)$$

$$Gy = -10 - 20 - 10 + 10 + 100 + 90 = 160$$

Step 3: Calculate the gradient magnitude

$$G = \sqrt{(Gx^2 + Gy^2)}$$

$$G = \sqrt{(240^2 + 160^2)} = \sqrt{(57{,}600 + 25{,}600)} = \sqrt{83{,}200}$$

$$G \approx 288.44$$

Step 4: Determine the edge direction

$$\theta = \tan^{-1}(Gy \div Gx) = \tan^{-1}(160 \div 240) \approx 33.69°$$

The gradient points approximately 33.69° from the positive horizontal axis. Since Gx is stronger than Gy, the intensity changes more strongly in the horizontal direction, indicating an approximately vertical edge with some diagonal variation.

Step 5: Applications of edge detection

1. Boundary detection: identifies object boundaries.

2. Image segmentation: helps separate objects from the background.

3. Object recognition: provides shape and boundary information.

4. Feature extraction: highlights important structures for further processing.

Result: Gx = 240, Gy = 160, gradient magnitude ≈ 288.44, and gradient angle ≈ 33.69°.

Summary of Final Answers

Question	Main result
Q1	Equalized values: 64, 64, 159, 159, 159, 223, 223, 255
Q2	Averaging filter centre = 29; median filter centre = 20
Q3	Laplacian response = 200; sharpened intensity = 300 before clipping
Q4	Foreground pixels = 5; background pixels = 4
Q5	Gx = 240; Gy = 160; magnitude ≈ 288.44; angle ≈ 33.69°